

Sabanci University

DSA210 Project Proposal Report

Kaan Toptaş

About My Project Idea

In this project, I will focus on the rating trends of famous directors across their filmography. My main research question is: When a director makes a long movie, does it negatively affect the rating of his/her next movie?

Source of the Data and How Will I Collect It?

My main data sources are the IMDb Movies Dataset and the TMDB 5000 Movies Dataset. Both will be downloaded from Kaggle.

Characteristics of the Data

The datasets contain movie titles, director names, runtimes, IMDb ratings, and production years. The IMDb dataset includes 1000 movies. The TMDB dataset includes 5000 movies. After merging on movie titles, approximately 450-500 common movies will be used for analysis.

About EDA and Hypothesis Test

After loading and merging the data, I will start with exploratory data analysis. First, I will perform single variable analysis to understand the distribution of movie runtimes and ratings, and to identify any outliers. Then I will proceed with multiple variable analysis to examine the correlation between movie runtimes and ratings.

For hypothesis testing, I will use an independent t-test. The null hypothesis is that there is no difference in rating change between directors whose previous movie was long (over 150 minutes) and those whose previous movie was short or medium (150 minutes or less). The alternative hypothesis is that directors who made a long movie experience a greater drop in their next movie's rating.

Expected Outcome

I expect to find a statistically significant negative effect. Directors who invest time and effort into long movies may experience fatigue, which could lower the quality of their next project.